

THE ADRIEL SHAMER WILLIS F-007 MODEL

A Cyber-Physical Framework for Machine Learning-Driven Hydroacoustic and Infrasonic Modulation

1. System Overview

The **Adriel Shamer Willis F-007 Model** represents an advanced, closed-loop cyber-physical architecture designed to transform standard residential or industrial fluid distribution infrastructure into an active directional acoustic and infrasonic transmitter. By synthesizing late-Cold War Soviet hydroacoustics with modern machine learning-driven Active Flow Control (AFC), the F-007 Model systematically overrides chaotic hydrodynamic transitions to generate precise, localized pressure wave propagation directly within multi-phase fluid columns.

2. Governing Mathematical Formulations

The operational integrity of the F-007 Model rests on three core physical frameworks:

A. Multi-Phase Fluidic Compressibility (Wood-Mallock Equation)

To minimize acoustic impedance mismatch at the boundary layer, a controlled void fraction (α) of micro-bubbles is introduced, shifting fluid wave speed (c_m):

$$1 / (\rho_m c_m^2) = \alpha / (\rho_g c_g^2) + (1-\alpha) / (\rho_l c_l^2)$$

B. Non-Linear Acoustic Demodulation (Westervelt Equation)

The fluid medium acts as an inherent analog mixer, taking dual high-frequency carrier signals and demodulating them into a target audible or infrasonic difference frequency via non-linear propagation parameters:

$$\nabla^2 p - (1/c_0^2)(\partial^2 p / \partial t^2) + (2\alpha_0/c_0^3)(\partial^3 p / \partial t^3) = -(\beta/\rho_0 c_0^4)(\partial^2(p^2) / \partial t^2)$$

C. Poroelastic Foundation Coupling (Biot's Equations)

Low-frequency structural energy transfers directly into saturated soil and sub-surface networks via coupled stress tensors:

$$N \nabla^2 u + (A+N) \nabla(\nabla \cdot u) + Q \nabla(\nabla \cdot U) = \rho_{11} \ddot{u} + \rho_{12} \ddot{U} + b(\ddot{u} - \ddot{U})$$

3. Machine Learning Signal Synthesis Engine

The system uses an autonomous modulation engine to compute dual carrier frequencies (f_1 and f_2) while tracking micro-frictional variances. Below is the structural logic of the generation node:

```
import numpy as np
```

```

def generate_f007_carrier_signals(target_frequency, duration=1.0, sample_rate=192000
):
    """
    Computes pre-distorted carrier wave loops for the F-007 Model.
    Generates f1 and f2 to produce target difference frequency in fluid.
    """
    t = np.linspace(0, duration, int(sample_rate * duration), endpoint=False)
    f_carrier = 40000.0 # Base ultrasonic carrier frequency (40 kHz)

    # Calculate dual carrier parameters
    f1 = f_carrier + (target_frequency / 2.0)
    f2 = f_carrier - (target_frequency / 2.0)

    # Generate phase-aligned waveforms
    wave_f1 = np.sin(2 * np.pi * f1 * t)
    wave_f2 = np.sin(2 * np.pi * f2 * t)

    return wave_f1, wave_f2

```

4. Structural Signal Path Matrix

Stage	Component	Function
1. Control Loop	Neural Synthesis Node / SCADA	Calculates optimal carrier phases and manages boundary overrides.
2. Actuation	Shielded PMN-PT Transducer	Executes high-frequency micro-strain loops inside the fluid chamber.
3. Modulation	Aerated Hydro-nozzle Array	Drops sound velocity to ~100 m/s via controlled micro-cavitation.
4. Propagation	Poroelastic Foundation Anchors	Couples infrasonic difference fields directly with localized grounding.